MAE 3128
Biomechanics-I

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Topics covered today:

1. Deformations due to Extension
    1. Anatomy of the long bone
    2. Biological motivations and applications
2. Displacement in vertically mounted, axially loaded members
3. In-class problems

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Summary of what's been covered in this course so far

- **Lectures 1, 2:** Introduced **Biomechanics** and reviewed **Statics** under rigid body assumptions
- **Lectures 3, 4:** Introduced **general concepts of stress and strain,** and **constitutive relationship using Hooke's law** for small-strain, solid-like materials.
- **Lectures 4, 5:** Shift focus to **canonical problems** in introductory biosolid mechanics

  - **For each problem,**
    - Choose coordinate systems
    - Calculate **stress components** by applying **equilibrium equations** (force, linear and angular momentum balances without inertia).
      - **Stress** is expressed in terms of **applied load and geometry**
      - **Material properties** appearing explicitly only in a few cases.
    - **Calculate strain or deformation**, depending on **load, geometry, and material properties.**

After the midterm exam the following **canonical problems** will be addressed:

- **Extension of rods and torsion of cylindrical structures**
- **Bending of beams**
- **Buckling of columns**

Biological Motivation: Why Study Bone Mechanics?

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)

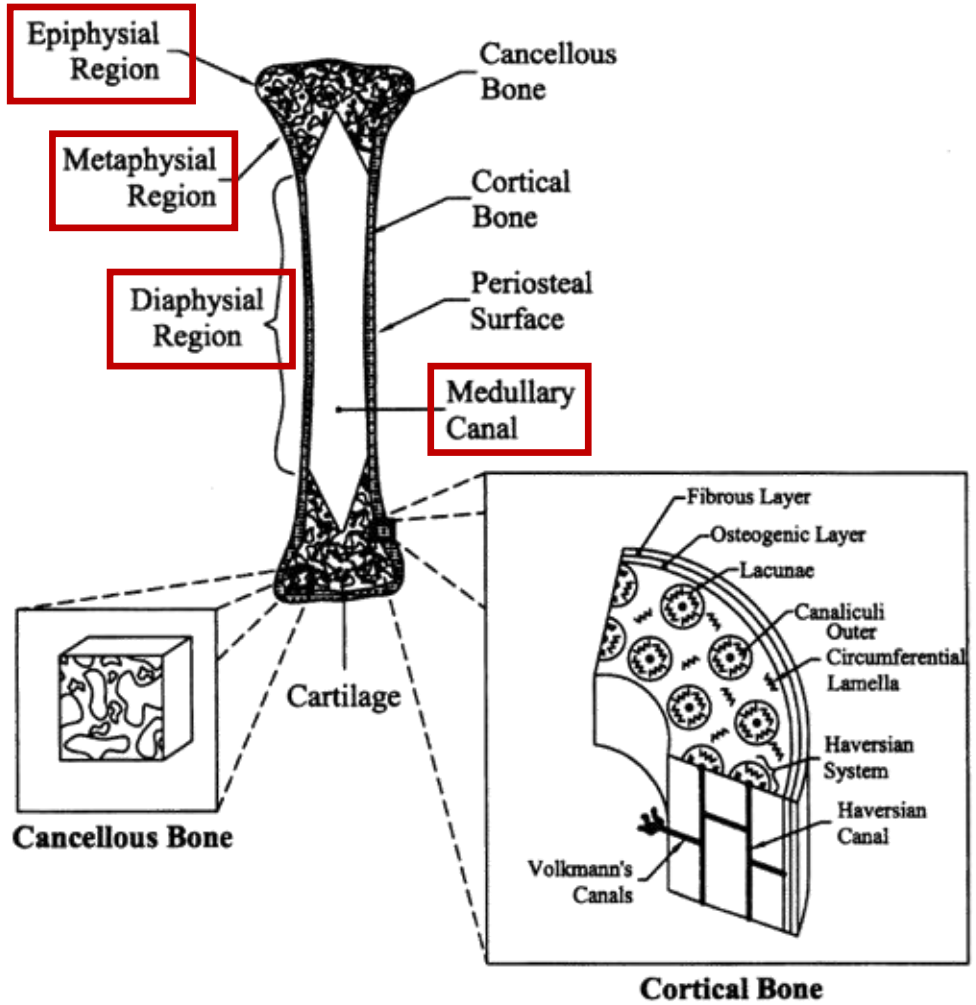
- Understanding how bone experiences and responds to mechanical loads is fundamental to bone mechanobiology.
- To determine whether osteocytes and surface cells can sense stress or strain, we must first **calculate the mechanical fields** (stress and strain) that bones actually experience.
- This requires integrating biological structure with mechanical analysis to understand how bones adapt, remodel, and respond to their mechanical environment.

Clinical and Research Relevance

- Laboratory mechanical testing protocols (isolated bone torsion)
- Understanding load distribution in physiological conditions
- Predicting bone response to altered mechanical environments
- Foundation for mechanobiological studies of cell sensing and adaptation

Structure of the Long Bone

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)

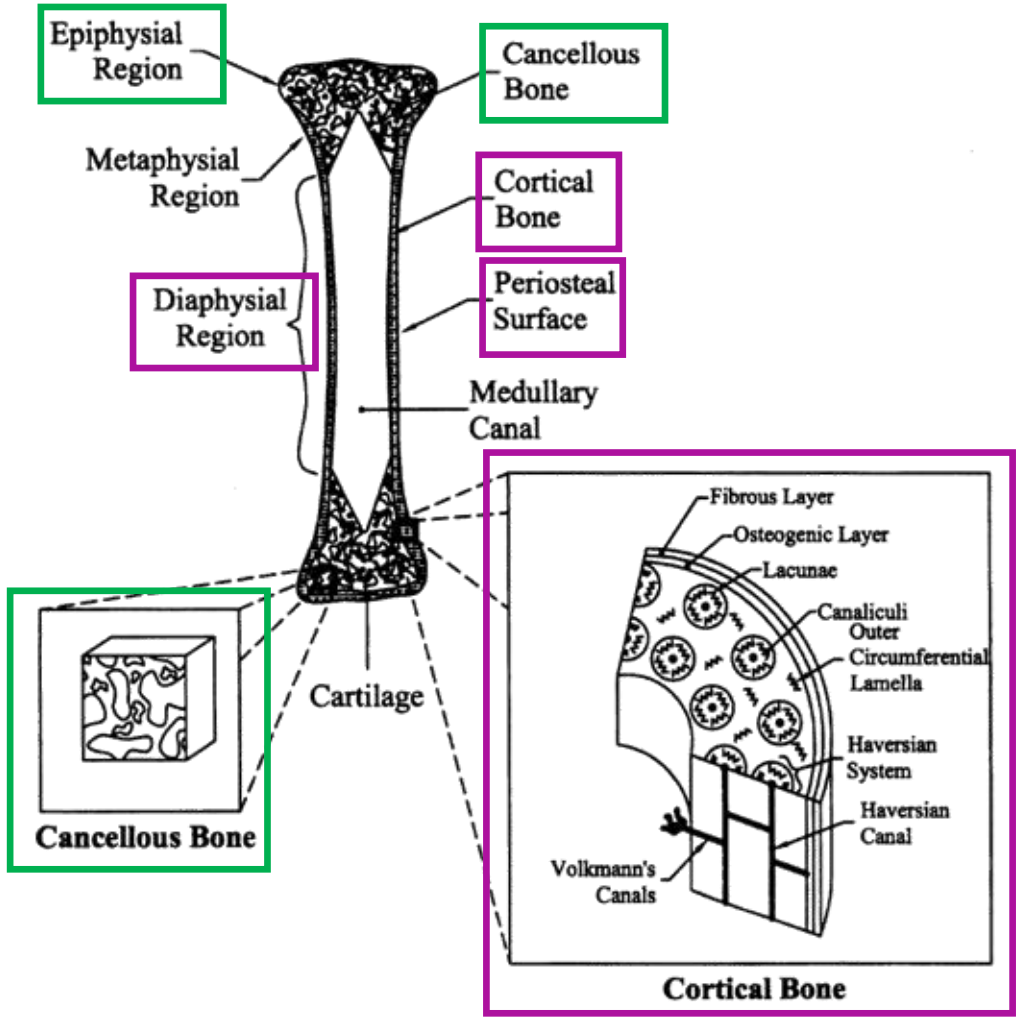


Gross Anatomical Regions

- **Diaphysial Region**: Central hollow shaft containing the **medullary canal** (bone marrow produces blood cells)
- **Epiphysial Region**: End caps of the bone
- **Metaphysial Region**: Transitional zones between shaft and ends

Two Primary Bone Tissue Classes

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)



Cancellous (Trabecular) Bone

- Found primarily in epiphysial regions
- Porous, spongy architecture (shown in inset diagram)

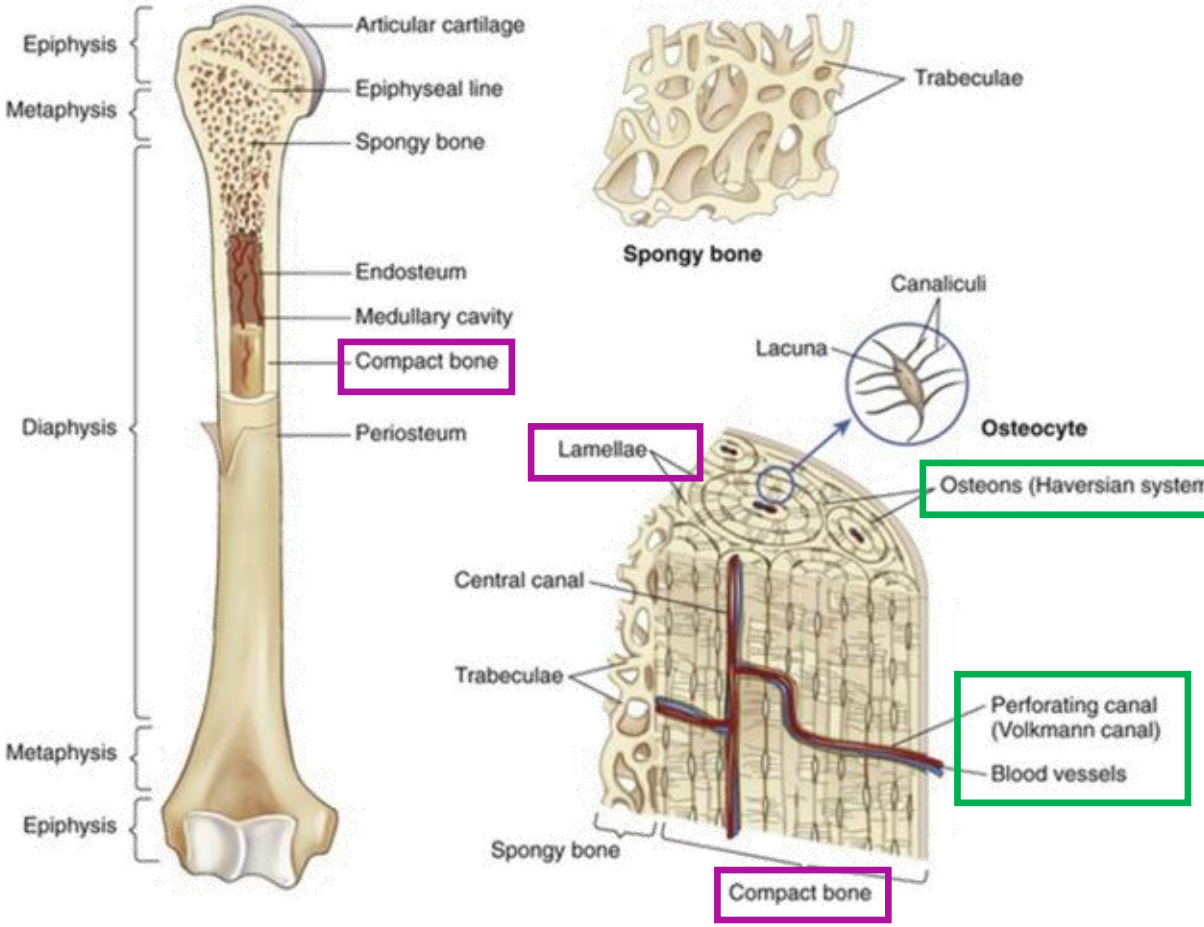
Cortical (Compact) Bone

- Constitutes outer portion of whole bone
- Forms majority of **diaphysial** wall
- Dense structure with specific gravity ~2
- Covered by **periosteum** (except in select regions)
  - Rich in collagen and fibroblasts
  - Underlying osteogenic layer contains active bone cells

Cortical Bone: Microstructure

References:

1. Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)
2. https://veteriankey.com/osteomyelitis/



Lamellar Bone

- Concentrically arranged layers (~20 µm thick each)
- Blood vessel networks between layers

Osteonal (Haversian) Bone

- **Cylindrical units (osteons):** ~200 µm diameter, ~2 cm long
- Central **Haversian canal** contains blood vessels
- **Volkmann's canals:** Radial channels connecting Haversian systems

Cortical Bone: Cellular Architecture

References:

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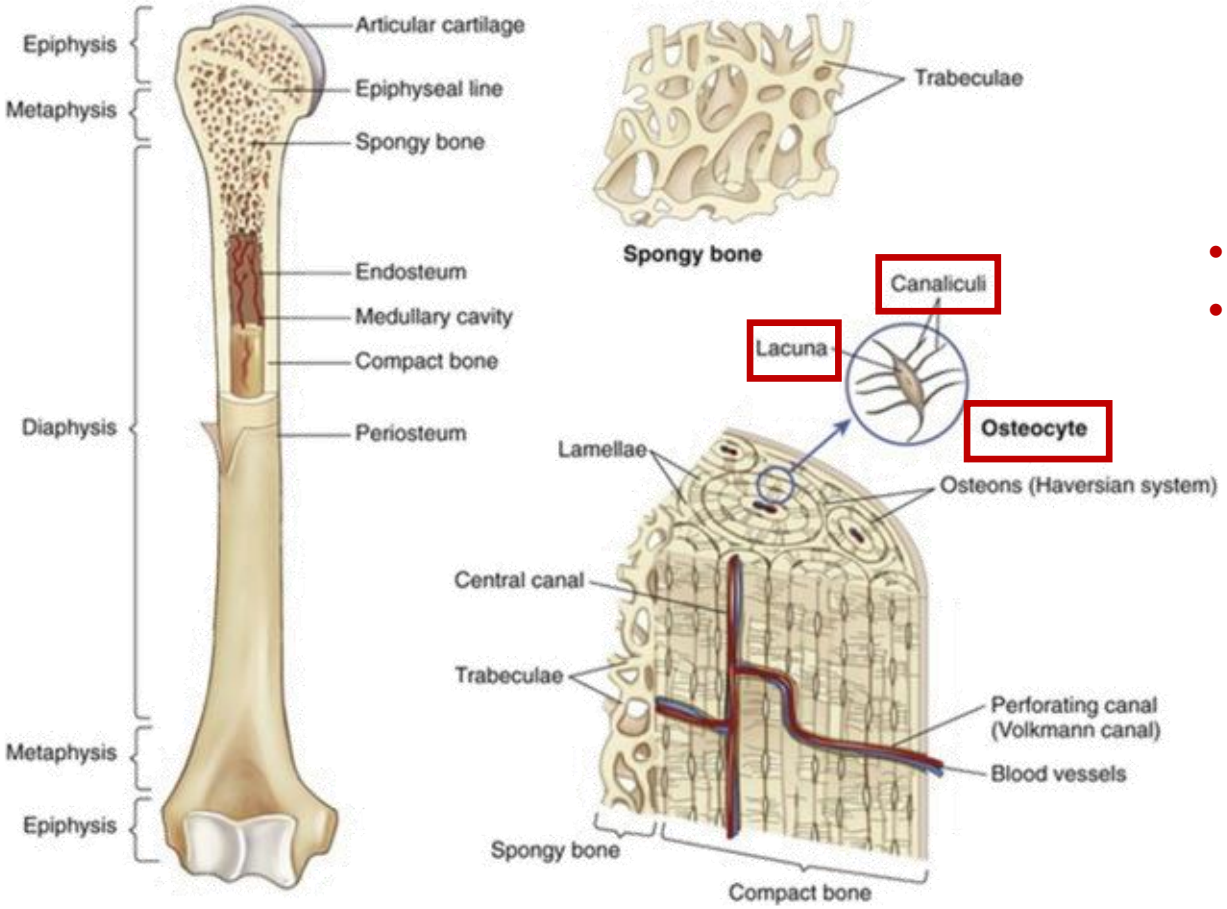

- **Lacunae:** Lenticular cavities housing **osteocytes (bone cells)**
- **Canaliculi:** Tubular passages radiating from lacunae
  - Enable nutrient exchange
  - Contribute to overall bone porosity

Modes and Sites of Bone Growth

- Bone grows by **appositional** deposition on existing, cell-laden surfaces, not by interstitial growth.
- Growth occurs at the **periosteal surface**, the **endosteal surface** lining the medullary canal, and on surfaces of tubular cavities and **trabeculae**.
- **Trabecular bone** grows via an increase in trabecular number and/or thickness.

Time Scales of Bone Adaptation

- Skeletal development spans years.
- Stress- or strain-mediated adaptation occurs over months to years.
- After injury (e.g., fracture), bone repair through new growth typically occurs within weeks to months.

References:

1. Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)
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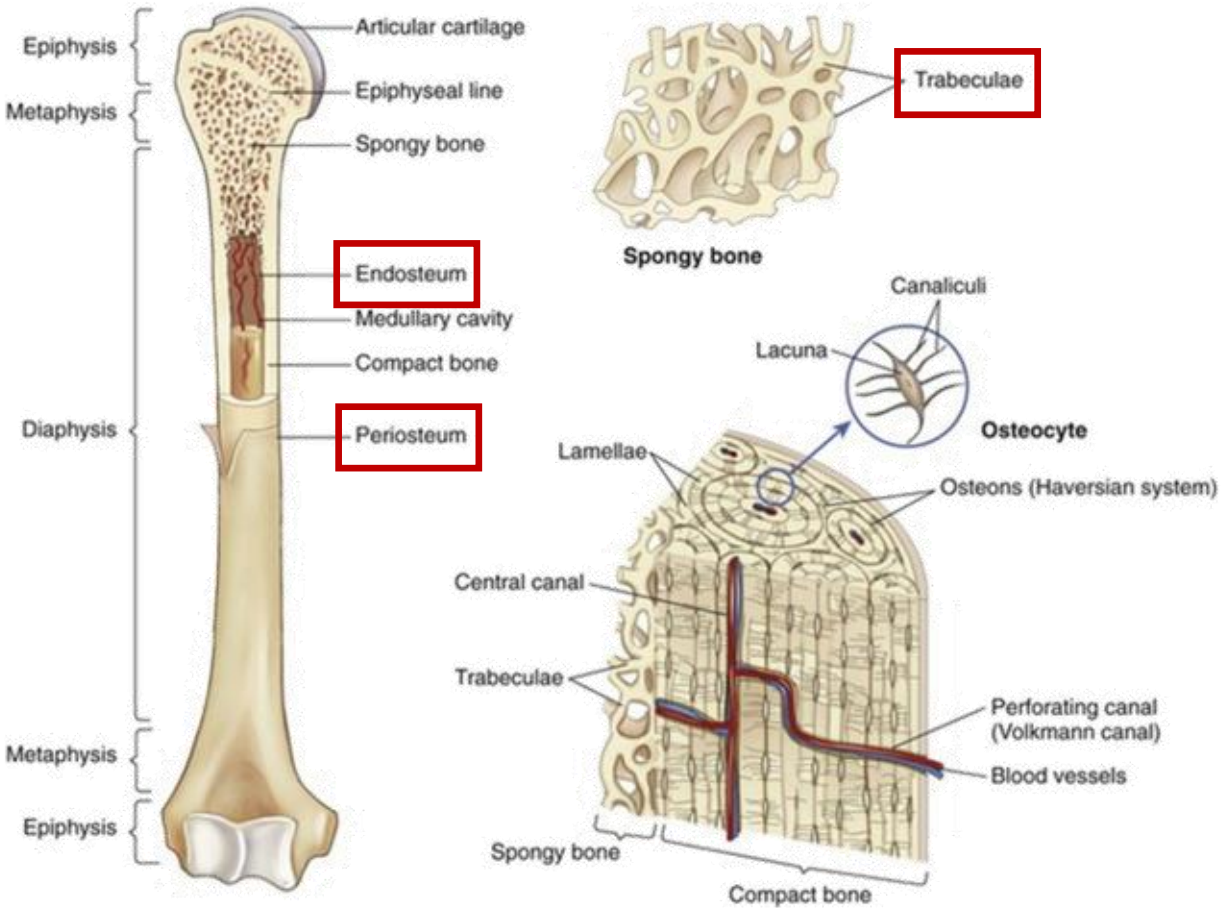

Bone composition, Cell types and Functions

References:

1. Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)
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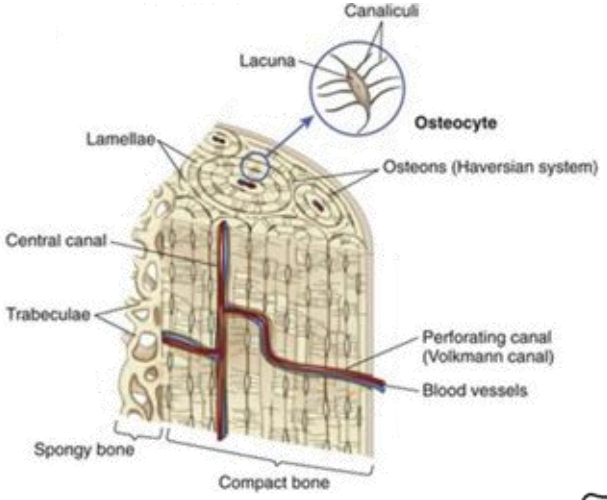
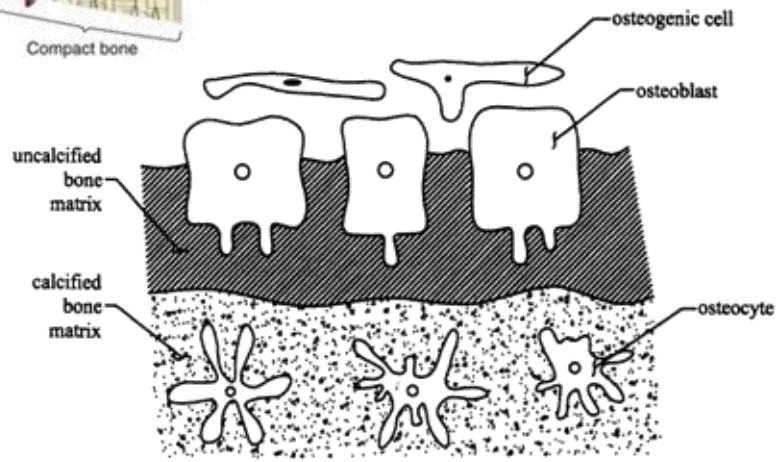
- Bone matrix is mainly type I collagen impregnated with hydroxyapatite $Ca_{10}\,(PO_4)_6\,(OH)_2$ giving high compressive strength.
- Despite the strong matrix, overall growth and remodeling are governed by bone cells.





Major Bone Cell Types and Functions

- **Osteoprogenitor cells**: Undifferentiated mesenchymal-derived cells on free surfaces; active during development and repair.
- **Osteoblasts**: Bone-forming cells that synthesize collagen and regulate mineral deposition and organization.
- **Osteoclasts**: Large multinucleated cells (20–100 µm) responsible for bone resorption.
- **Osteocytes**: Mature bone cells derived from osteoblasts, residing in lacunae; they no longer divide once encased in calcified matrix.

Osteocyte Network and Cell Communication

- Osteocytes extend processes through canaliculi, forming gap junctions with neighboring osteocytes.
- Through this network they likely participate in regulating osteoblast and osteoclast activity and thus control remodeling.

Mechanical Analysis of Bone

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)

Requirements for Stress/Strain Calculation

- To calculate stress or strain fields in bone, three key inputs are needed:
  - **Geometry**: Bone shape and structural architecture
  - **Material properties**: Elastic moduli, density, porosity
  - **Applied loads**: Compression, tension, bending, torsion

Common Loading Scenarios for Practical Analysis:

- **Axial compression**: Weight-bearing loads
- **Torsion**: Twisting moments (e.g., femur rotation)
- **Bending**: Transverse or multiplanar loads
- Focus on diaphysial (shaft) region
- Cortical bone only

Simple Models (LEHI Approximation)

- Linear, Elastic, Homogeneous, Isotropic behavior
- Effective properties: $E \approx 15$ GPa, $\nu \approx 0.33$
- Suitable for bulk stress analysis under moderate loads

Advanced Models (Current Research)

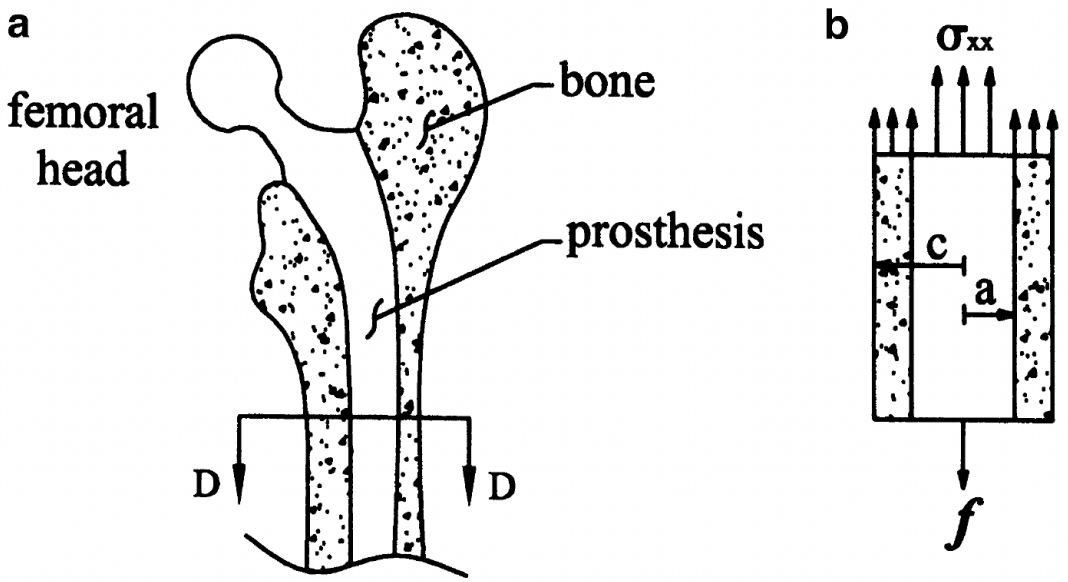
- Fine trabecular architecture modeling
- Porosity effects on material behavior
- Solid-fluid coupling (blood/bone fluid flow under load)
- Accounts for internal fluid dynamics in response to mechanical loading

Clinical Application: Prosthetic Hip Implants

Reference: Humphrey, J. D., & O'Rourke, S. L. (2015). *An introduction to biomechanics: Solids and fluids, analysis and design* (2nd ed.)

Clinical Context

- ~120,000 artificial hips implanted annually in the US
- Purpose: relieve pain and restore ambulatory motion
- Common cause: osteoporotic fractures in elderly women

Biomechanical Analysis

- **Model:** axial loading of bone-metal interface
- **LEHI assumptions:** simplified homogeneous behavior
- **Key question:** How does an implant redistribute stress in bone